



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

1. Data Collection using SpaceX API and web scraping from Wikipedia
2. Exploratory Data Analysis (EDA) with SQL
3. Data Visualization using Folium and Plotly Dash to create interactive visual analytics and dashboards
4. Model Building and Evaluation using Logistic Regression, Support Vector Machine (SVM), Decision Trees, and K Nearest Neighbors,

GitHub URL:

<https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/tree/main>

Summary of all results

1. Data Insights and Visualization
 - Identified factors influencing Falcon 9 first stage landings.
 - Visualized geographical patterns and success rates.
2. Machine Learning Model Performance:
 - Decision Tree: 94.44% accuracy.
 - Logistic Regression, SVM, and K-Nearest Neighbors: 83.33% accuracy.
3. Key Findings:
 - Launch site and payload mass influences successful landing
 - Decision Tree model is the most effective machine learning model

Introduction

- Project background and context
 - In this capstone project, we will predict if the Falcon 9 first stage will land successfully. SpaceX advertises that Falcon 9 rocket are significantly cheaper than other providers due to the savings from reusing the first stage of the rocket.
- Problems you want to find answers
 - Determine the cost of the launch if the first stage lands.
 - How to predict the landing outcome using the machine learning models.
 - Which machines learning models performs the best in predicting the landing success of Falcon 9.

Section 1

Methodology

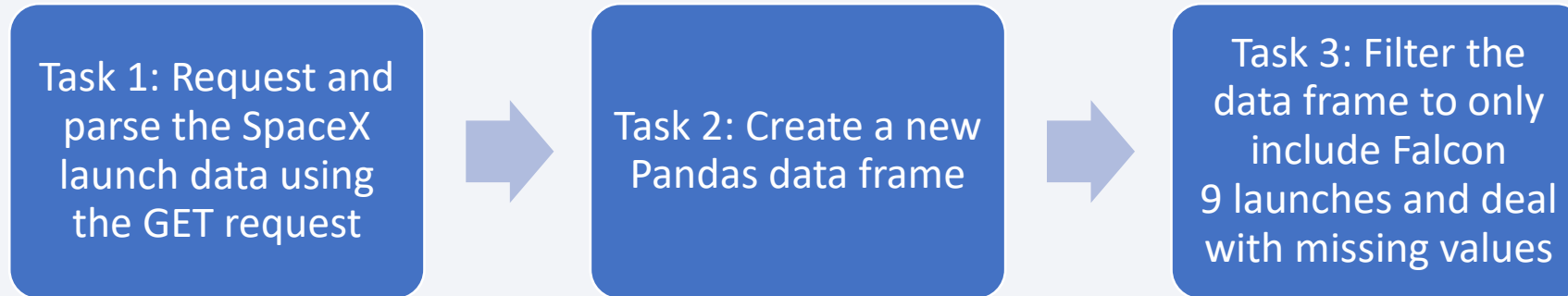
Methodology

Executive Summary

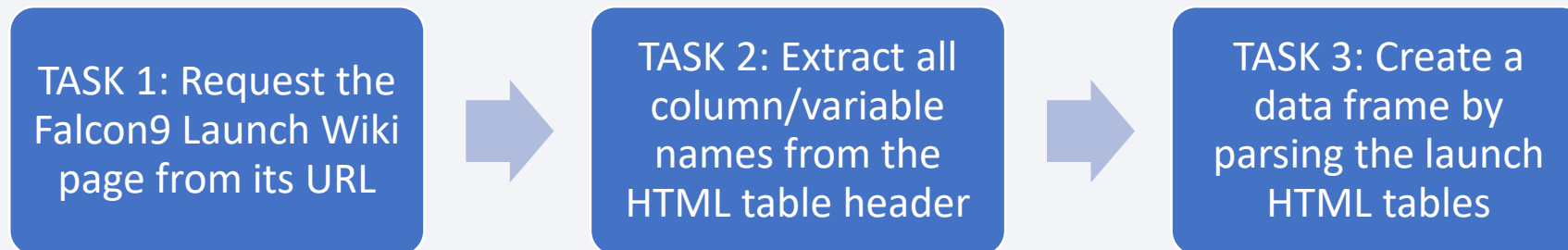
- Data collection methodology:
 - Data was extracted via SpaceX REST API endpoints which gives us data about launches, rocket used, payload delivered, launch specifications, landing specifications, and landing outcome.
 - Data was extracted from the SpaceX API and web scraped Falcon9 launch data from SpaceX Wiki page as HTML tables.
- Perform data wrangling
- Perform Exploratory Data Analysis (EDA) using SQL to visualized and understand data to find patterns and relationships.
- Perform interactive visual analytics using Folium map and Plotly Dash dashboard
- Perform predictive analysis using classification models

Data Collection

- Step 1: Request and parse SpaceX launch data using the API



- Step 2: Web scrape Falcon 9 launch records from Wikipedia



Data Collection – SpaceX API

- Task 1: Request and parse the SpaceX launch data using the GET request
 - `spacex_url="https://api.spacexdata.com/v4/launches/past"`
- Task 2: Create a new Pandas data frame
- Task 3: Filter the data frame to only include Falcon 9 launches and deal with missing values
- GitHub URL:
<https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

Import Libraries and Define Auxiliary Functions

Initiate API Request to fetch launch data (`requests.get`)

Parse JSON
Response (`response.json()`)

Extract relevant columns (rocket, payloads, launchpad, cores, flight number, date.)

Store Data in DataFrame (`pandas.DataFrame`) and save as CSV

Data Collection - Scraping

- TASK 1: Request the Falcon9 Launch Wiki page from its URL
 - TASK 2: Extract all column/variable names from the HTML table header
 - TASK 3: Create a data frame by parsing the launch HTML tables
-
- GitHub URL:
<https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/jupyter-labs-webscraping.ipynb>

Request the Falcon9 Launch Wiki page (requests.get)

Create BeautifulSoup object

Extract Launch Records stored in a HTML table

Create a data frame by parsing the launch HTML tables (pandas.DataFrame)

Data Wrangling

- The SpaceX Falcon 9 launch dataset was loaded into a Pandas DataFrame for data wrangling and exploratory data analysis. The data was checked for missing values, data types, launch site counts, orbit frequencies, and mission outcome categories. Landing outcomes were labeled 1 for successful landings and 0 for unsuccessful landings. Final dataset was saved as a new CSV file to be used later in machine learning modeling.
- GitHub URL: <https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Raw SpaceX Data



Data Inspection



Missing Value Check



Exploratory Data Analysis



Outcome Classification



Processed Dataset for Machine Learning

EDA with Data Visualization

- Created a scatter plot for FlightNumber vs PayloadMass, FlightNumber vs LaunchSite, Payload Mass vs Launch Site, FlightNumber vs Orbit Type, Payload Mass and Orbit Type
- Created a bar chart for Success Rate vs Orbit Type
- Created line chart for Success Rate vs Year which shows data trends over time
- GitHub URL: <https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/edadataviz.ipynb>

EDA with SQL

- Performed the following SQL queries:
 - Task 1: Display the names of the unique launch sites in the space mission
 - Task 2: Display 5 records where launch sites begin with the string 'CCA'
 - Task 3: Display the total payload mass carried by boosters launched by NASA (CRS)
 - Task 4: Display average payload mass carried by booster version F9 v1.1
 - Task 5: List the date when the first successful landing outcome in ground pad was achieved.
 - Task 6: List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - Task 7: List the total number of successful and failure mission outcomes
 - Task 8: List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
 - Task 9: List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
 - Task 10: Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- GitHub URL: https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Summary

- TASK 1: Mark all launch sites on a map using markers
- TASK 2: Mark the success/failed launches for each site on the map using markers
- TASK 3: Calculate the distances between a launch site to its proximities using lines

Explanation for adding objects: Markers of launches were used to calculate distances from other point of interests such as coastlines, railways, etc.

GitHub URL: https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/lab_jupyterlaunch_site_location.ipynb

Build a Dashboard with Plotly Dash

Plotly Dash dashboard contains a interactive pie chart and scatterplot that updates based on launch site selection and payload range slider. The pie chart shows total successful launches by launch site and the scatter plot shows payload vs launch outcomes by sites.

- TASK 1: Add a Launch Site Drop-down Input Component
- TASK 2: Add a callback function to render success-pie-chart based on selected site dropdown
- TASK 3: Add a Range Slider to Select Payload
- TASK 4: Add a callback function to render the success-payload-scatter-chart scatter plot

GitHub URL: <https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

- This process used standardized data to train-test splits in order to evaluate the following models: SVM, Decision Trees, and KNN. GridSearchCV, cross-validation, and metrics such as accuracy, precision, recall, and F1-score were used to tune and select the best-performing model.
- GitHub URL:
https://github.com/samuelsong-ds/ibm-applied-data-science-capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Preprocessing: Standardize data, split data into training and test sets for model validation

Train the model and perform grid search

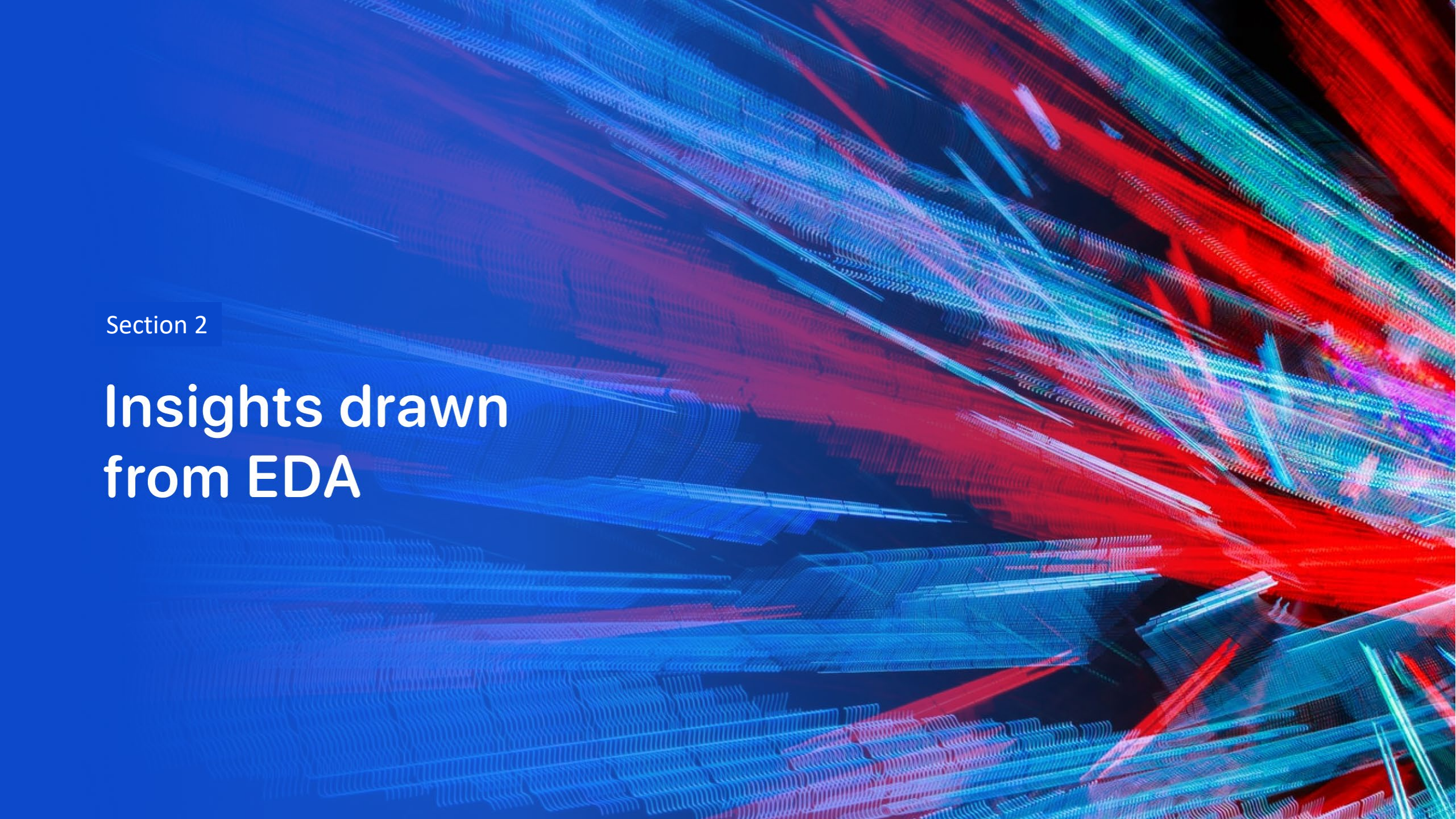
Use the hyperparameters to determine the best model

Evaluate the model and select the best performing model

Output Confusion Matrix

Results

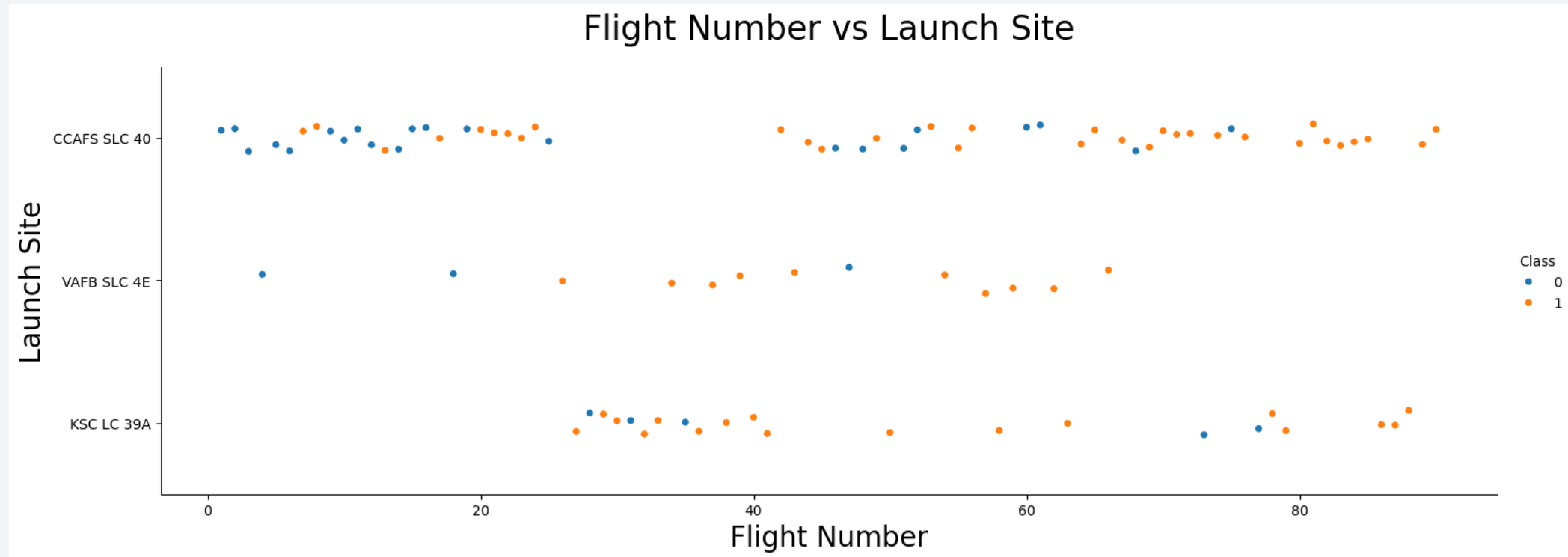
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

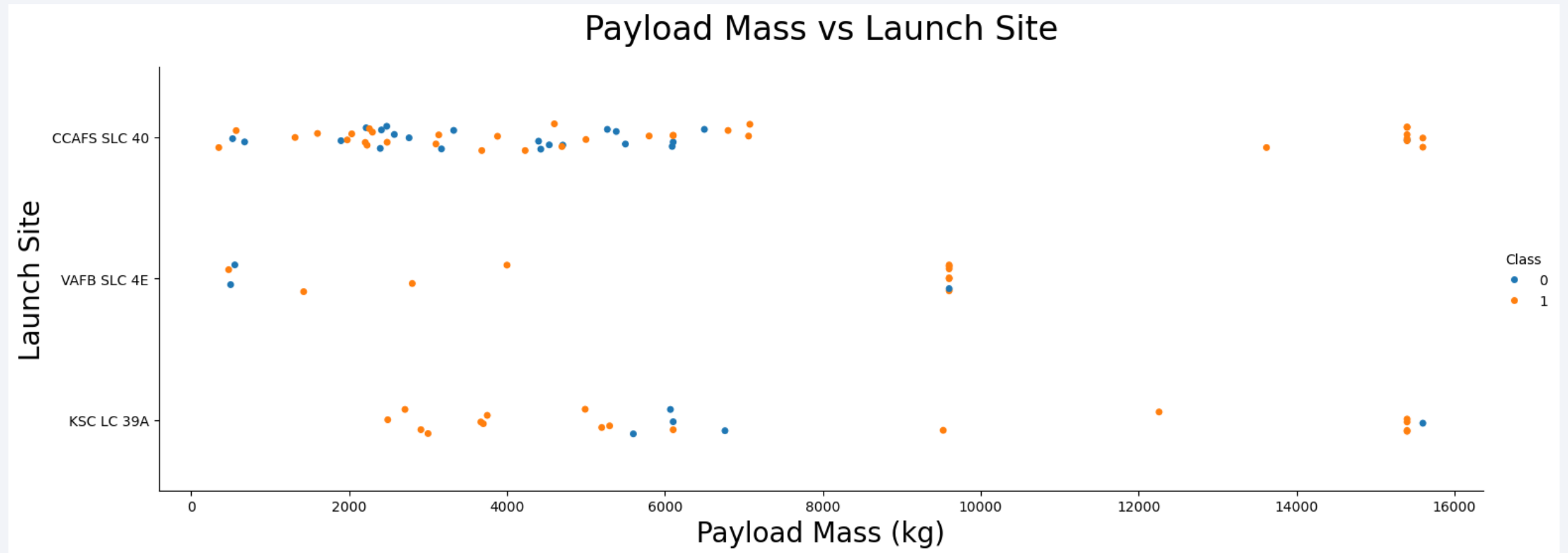
Insights drawn from EDA

Flight Number vs. Launch Site



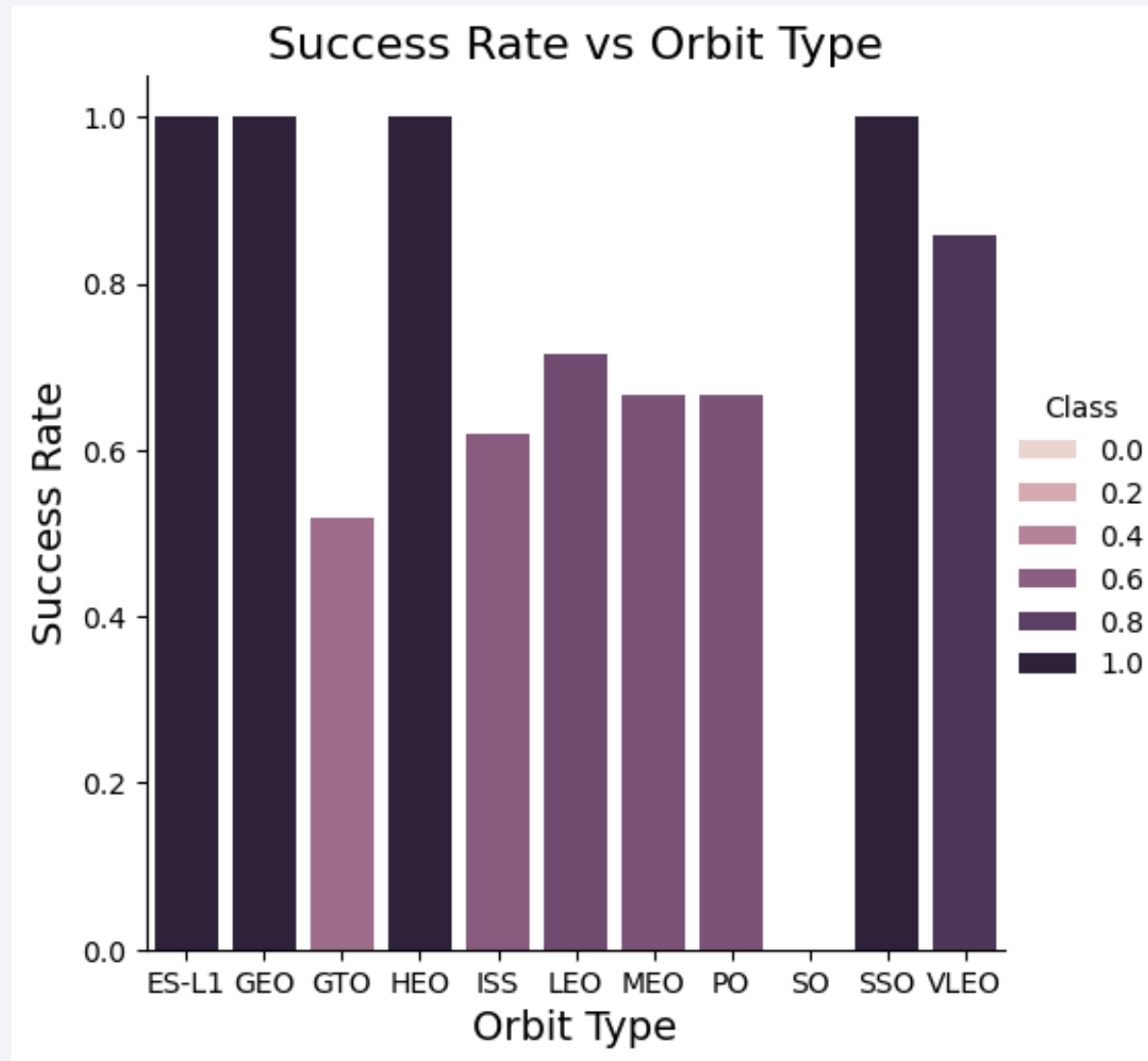
- Launch sites CCAFS SLC 40 and KSC LC 39A have a mix of successful and unsuccessful landings which points to other factors influencing the launch outcome other than the launch site itself.

Payload vs. Launch Site



- CCAFS SLC 40 had the most launches with the payload below 8,000 kg.
- VAFB SLC 4E did not have any launches with the payload over 10,000 kg.

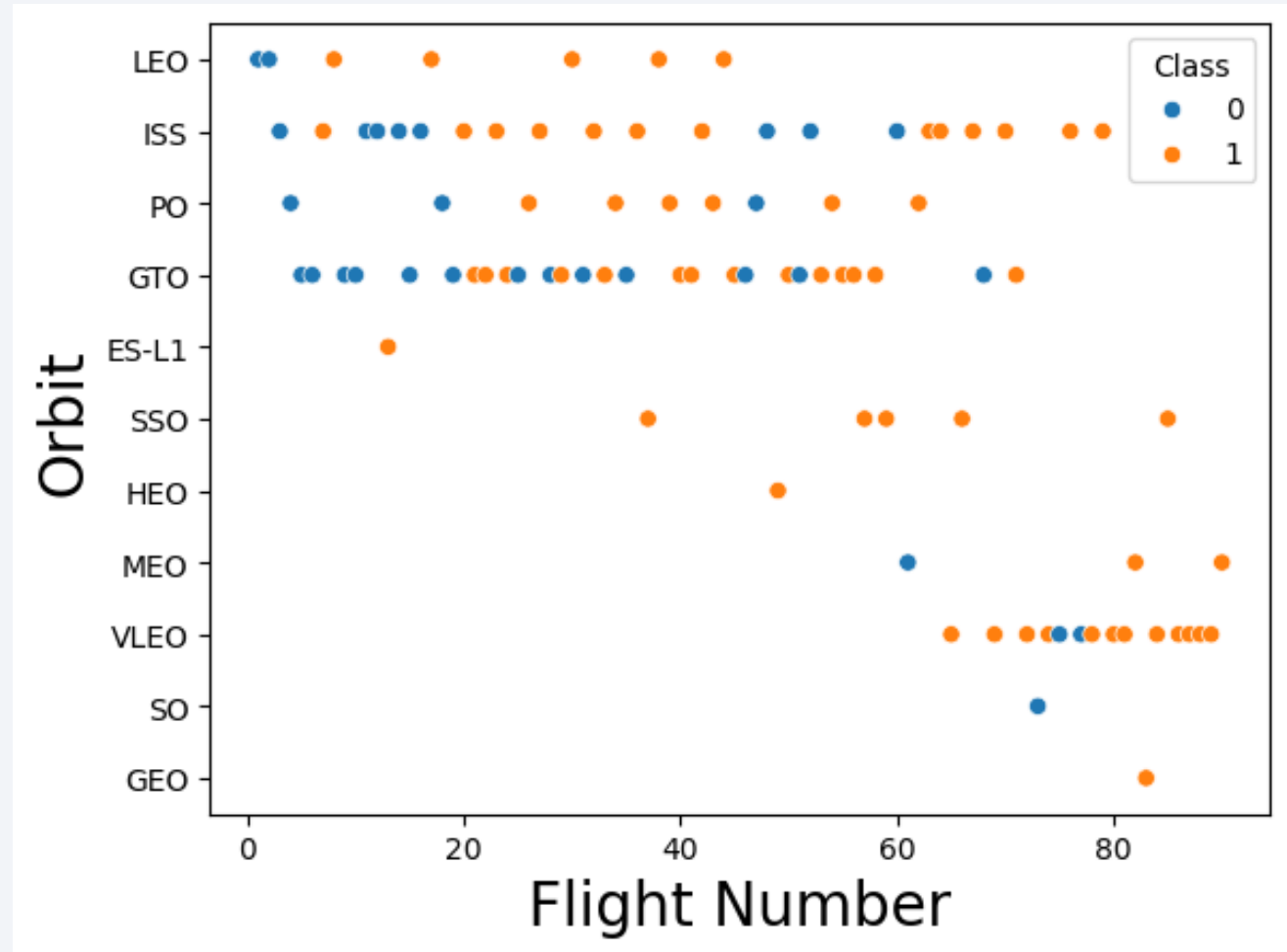
Success Rate vs. Orbit Type



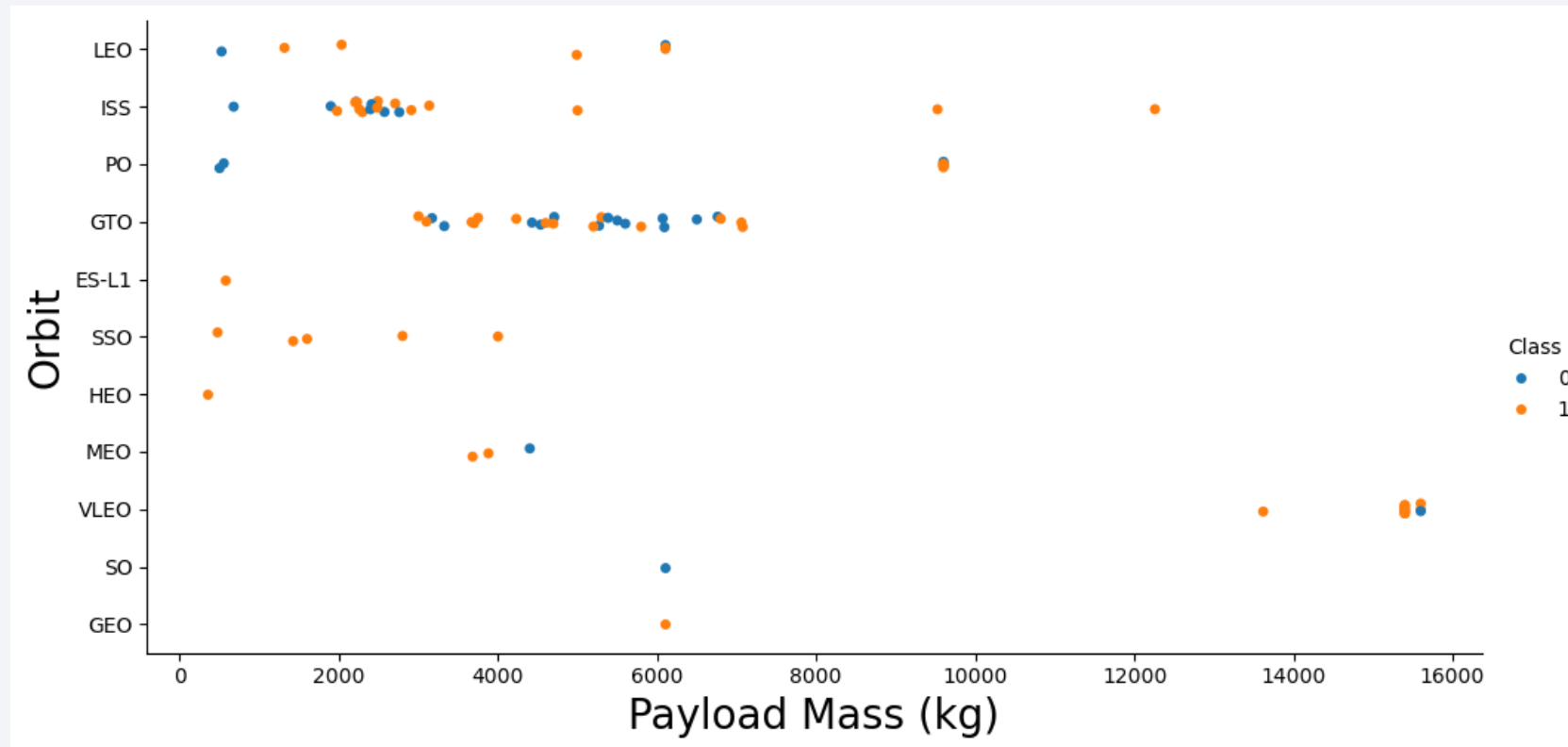
- Missions to ES-L1, GEO, HEO, SSO, and VLEO had the highest success rate over 80% indicating the orbits are reliable for successful first stage landing.
- Mission to GTO had the lowest success rate.

Flight Number vs. Orbit Type

- Success rate of Falcon 9 launches improved with higher flight numbers.
- Early flights to ISS and GTO orbits had mixed outcomes but success rate improved in the recent flights.



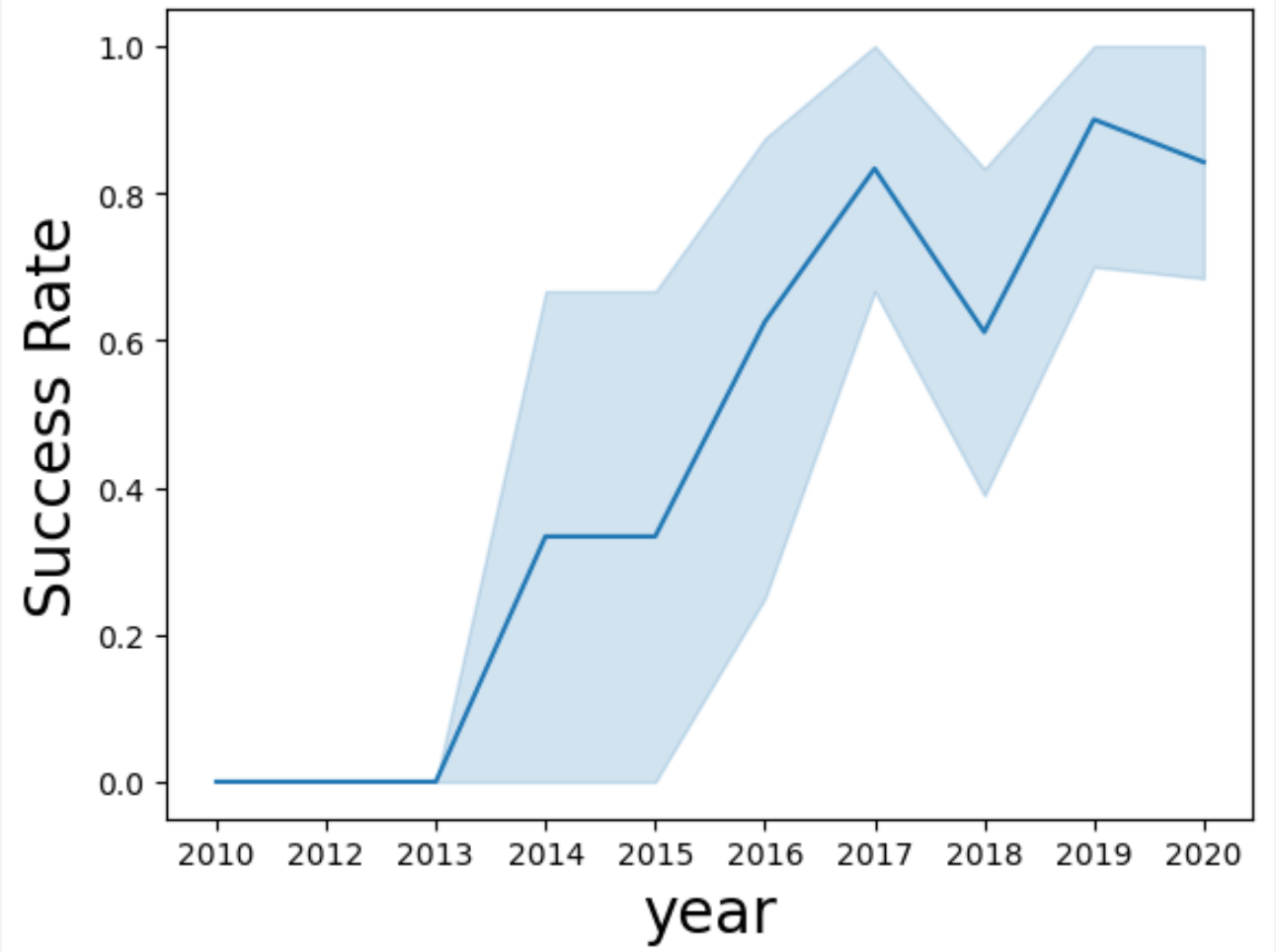
Payload vs. Orbit Type



- Most of the payload were below 8,000 kg
- More launches used ISS and GTO orbits

Launch Success Yearly Trend

- Success rate has been increasing since 2013.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[32]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[32]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Short Explanation: Display the names of the unique launch sites in the space mission

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[33]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[33]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Short Explanation: Display 5 records where launch sites begin with the string 'CCA'

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[34]: %sql SELECT SUM("Payload_Mass__kg_") AS Total_Payload_Mass FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[34]: Total_Payload_Mass
```

```
45596
```

Short Explanation: Display the total payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[35]: %sql SELECT AVG("Payload_Mass__kg_") AS Avg_Payload_Mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

```
[35]: Avg_Payload_Mass
```

```
2928.4
```

Short Explanation: Display average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[36]: %sql SELECT MIN("Date") AS First_Successful_Landing_Outcome FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
[36]: First_Successful_Landing_Outcome
```

```
2015-12-22
```

Short Explanation: List the date when the first successful landing outcome in ground pad was achieved.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[37]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Mission_Outcome" = 'Success' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000
```

```
* sqlite:///my_data1.db  
Done.
```

```
[37]: Booster_Version
```

F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1014
F9 v1.1 B1016
F9 FT B1020
F9 FT B1022
F9 FT B1026
F9 FT B1030

Short Explanation: List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[38]: %sql SELECT "Mission_Outcome", COUNT(*) AS total_number FROM SPACEXTABLE GROUP BY "Mission_Outcom";
```

```
* sqlite:///my_data1.db
```

Done.

```
[38]: Mission_Outcome  total_number
```

Mission_Outcome	total_number
Success	101

Short Explanation: List the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[39]: %sql SELECT "Booster_Version", "Payload_Mass" FROM SPACEXTABLE WHERE "Payload_Mass" = (SELECT MAX("Payload_Mass") FROM SPACEXTABLE);  
* sqlite:///my_data1.db  
Done.
```

```
[39]: Booster_Version  "Payload_Mass"  
-----  
F9 v1.0 B0003      Payload_Mass  
F9 v1.0 B0004      Payload_Mass  
F9 v1.0 B0005      Payload_Mass  
F9 v1.0 B0006      Payload_Mass  
F9 v1.0 B0007      Payload_Mass  
F9 v1.1 B1003      Payload_Mass
```

Short Explanation: List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[41]: %sql SELECT strftime('%m', Date) AS month_number, strftime('%Y-%m-%d', Date) AS Date, Booster_Version, Launch_Site, Landing_Outcome FROM SPAC
```

```
* sqlite:///my_data1.db  
Done.
```

```
[41]:
```

	month_number	Date	Booster_Version	Launch_Site	Landing_Outcome
	01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
	04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Short Explanation: List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[42]: %sql SELECT Landing_Outcome, COUNT(*) AS Landing_Outcomes FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome
```

```
* sqlite:///my_data1.db  
Done.
```

```
[42]:
```

Landing_Outcome	Landing_Outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Short Explanation: Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

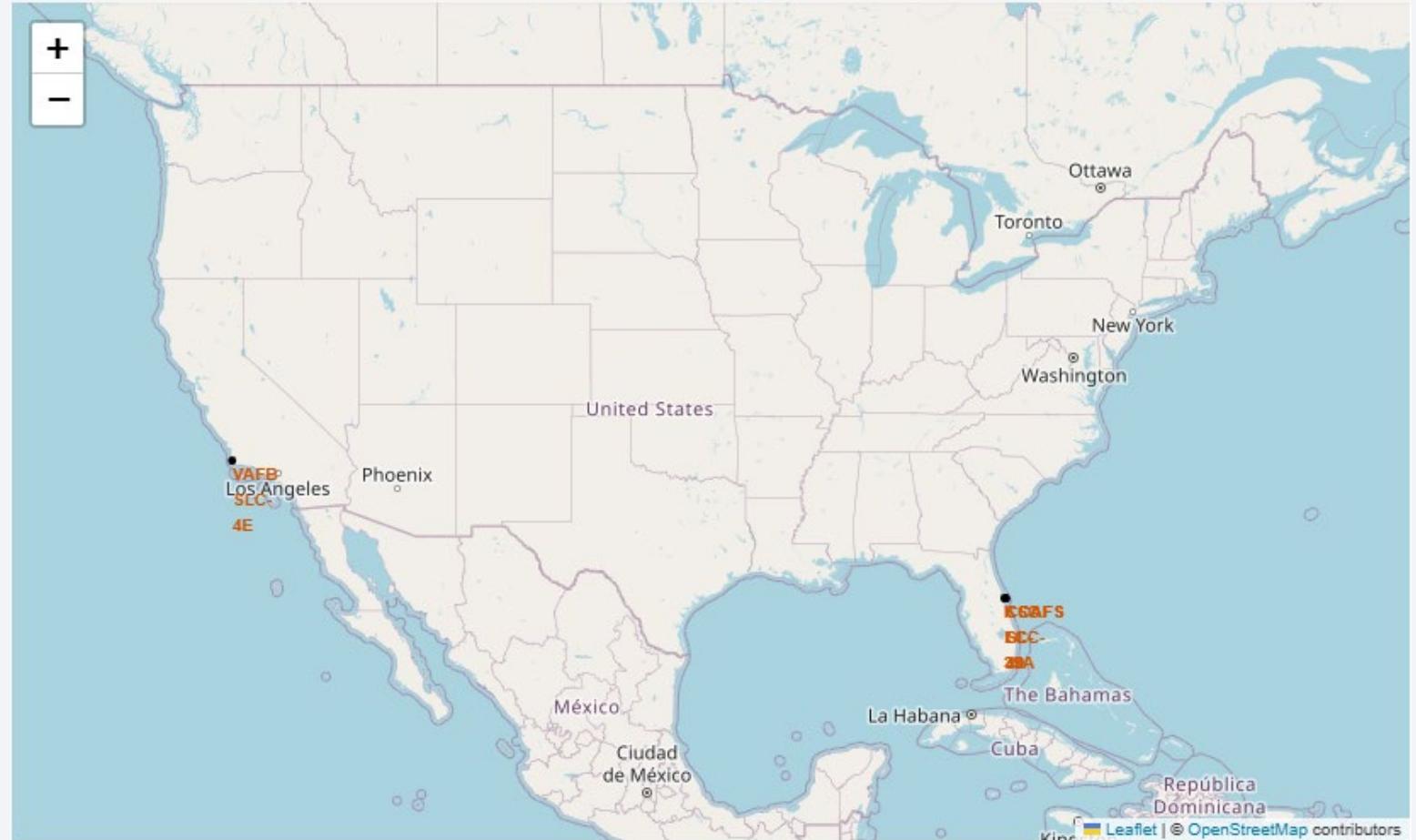
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is visible, showing clouds and city lights at night. The text "Section 3" is overlaid on the blue background.

Section 3

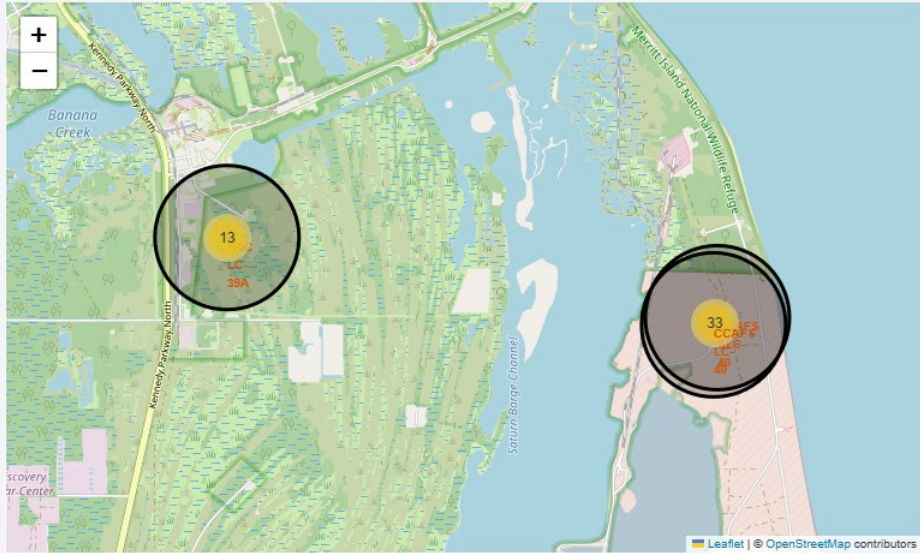
Launch Sites Proximities Analysis

TASK 1: Mark all launch sites on a map

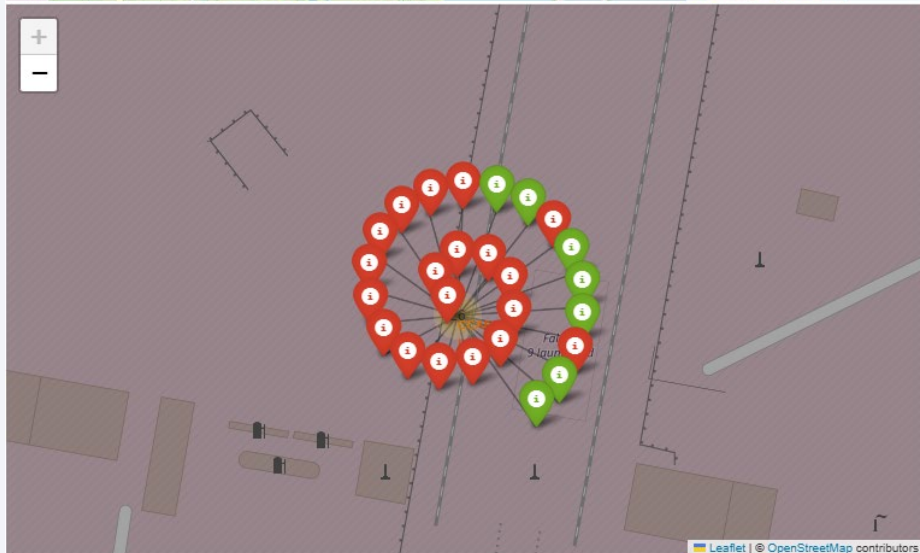
- Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



TASK 2: Mark the success/failed launches for each site on the map



- This folium map shows the successful (7 green markers) and failed launch outcomes (19 red markers) for launch site CCAFS LC-40.



Task 3: Calculate the distances between a launch site to its proximities

```
[25]: # find coordinate of the closet coastline
# e.g.,: Lat: 28.56367 Lon: -80.57163
# distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

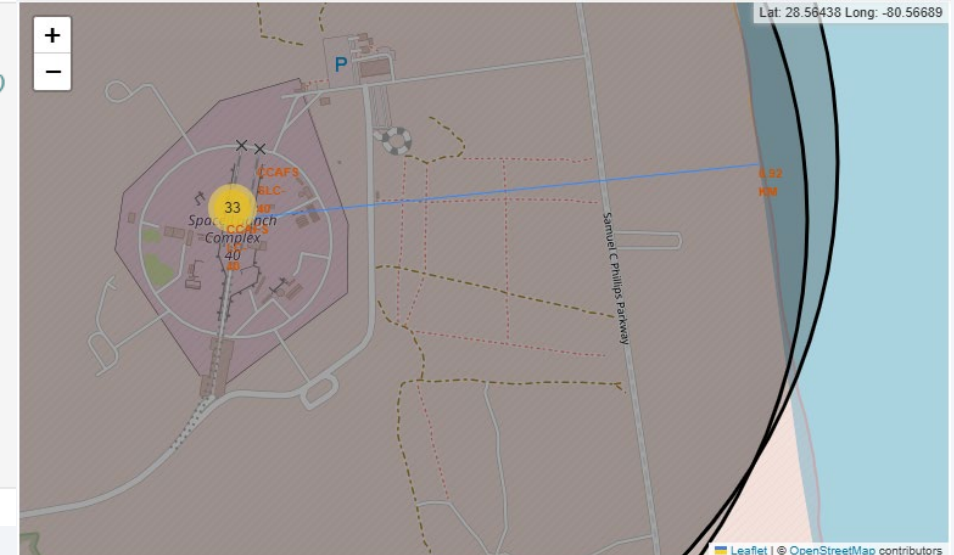
# Assume these are the coordinates of the closest coastline obtained from MousePosition
coastline_lat = 28.56317
coastline_lon = -80.56797

# Coordinates of the launch site (replace these with the actual launch site coordinates)
launch_site_lat = launch_sites_df.iloc[0]['Lat']
launch_site_lon = launch_sites_df.iloc[0]['Long']

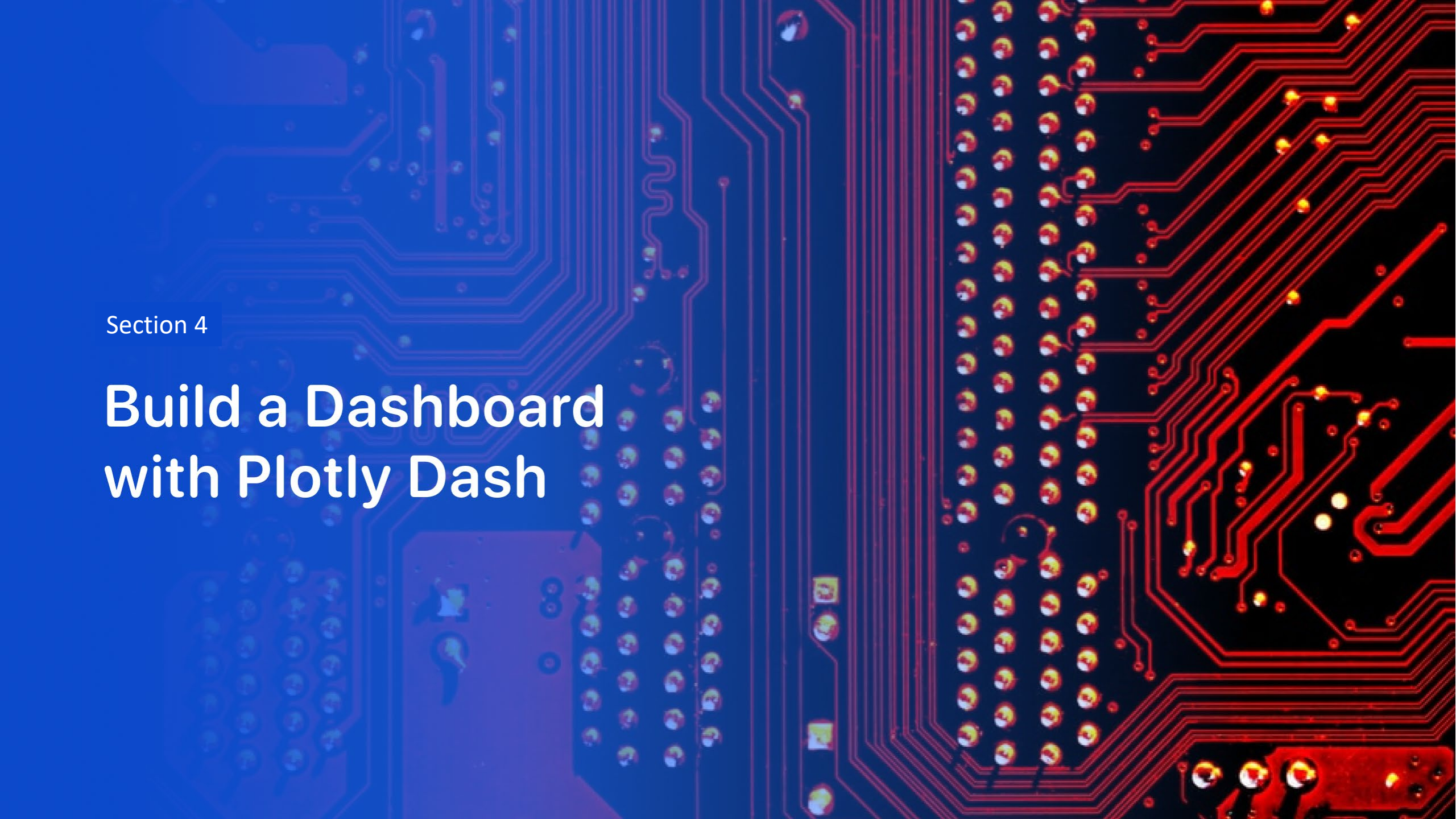
# Calculate the distance between the launch site and the coastline
distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

print(f"Distance to the closest coastline: {distance_coastline:.2f} km")

Distance to the closest coastline: 0.92 km
```



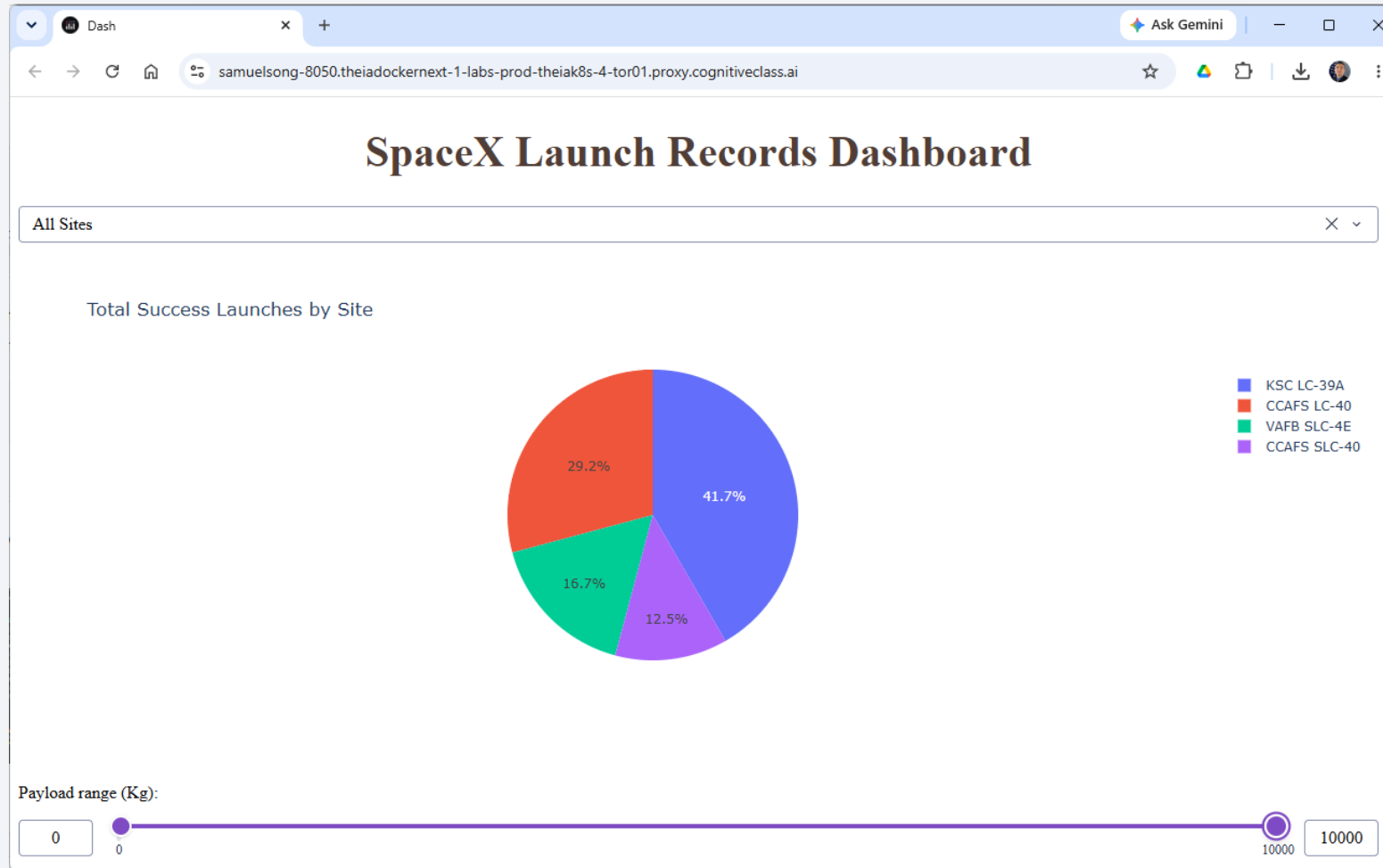
- This folium map shows CCAFS SLC-40 launch site to its closest coastline. Distance is 0.92 km from the coastline with a blue line symbolizing the distance. Close proximity to the coastline is typical for launch sites for safe-recovery and safety.



Section 4

Build a Dashboard with Plotly Dash

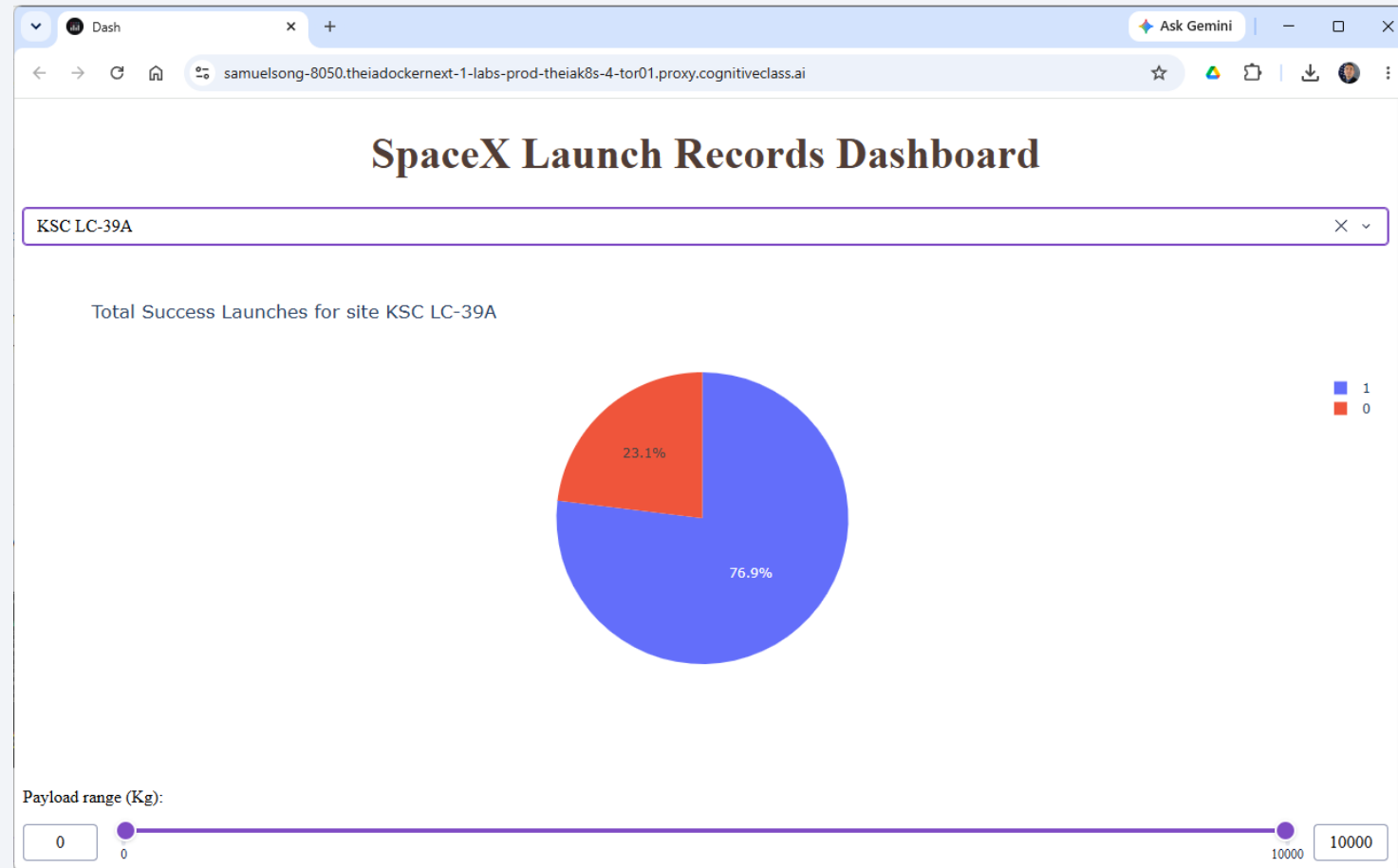
Total Successful Launches by Site



All Sites

- KSC LC-39A: 41.7%
- CCAFS LC-40: 29.2%
- VAFB SLC-4E: 16.7%
- CCAFS SLC-40: 12.5%
- KSC LC-39A has the highest number of successful launches making it the more reliable site.

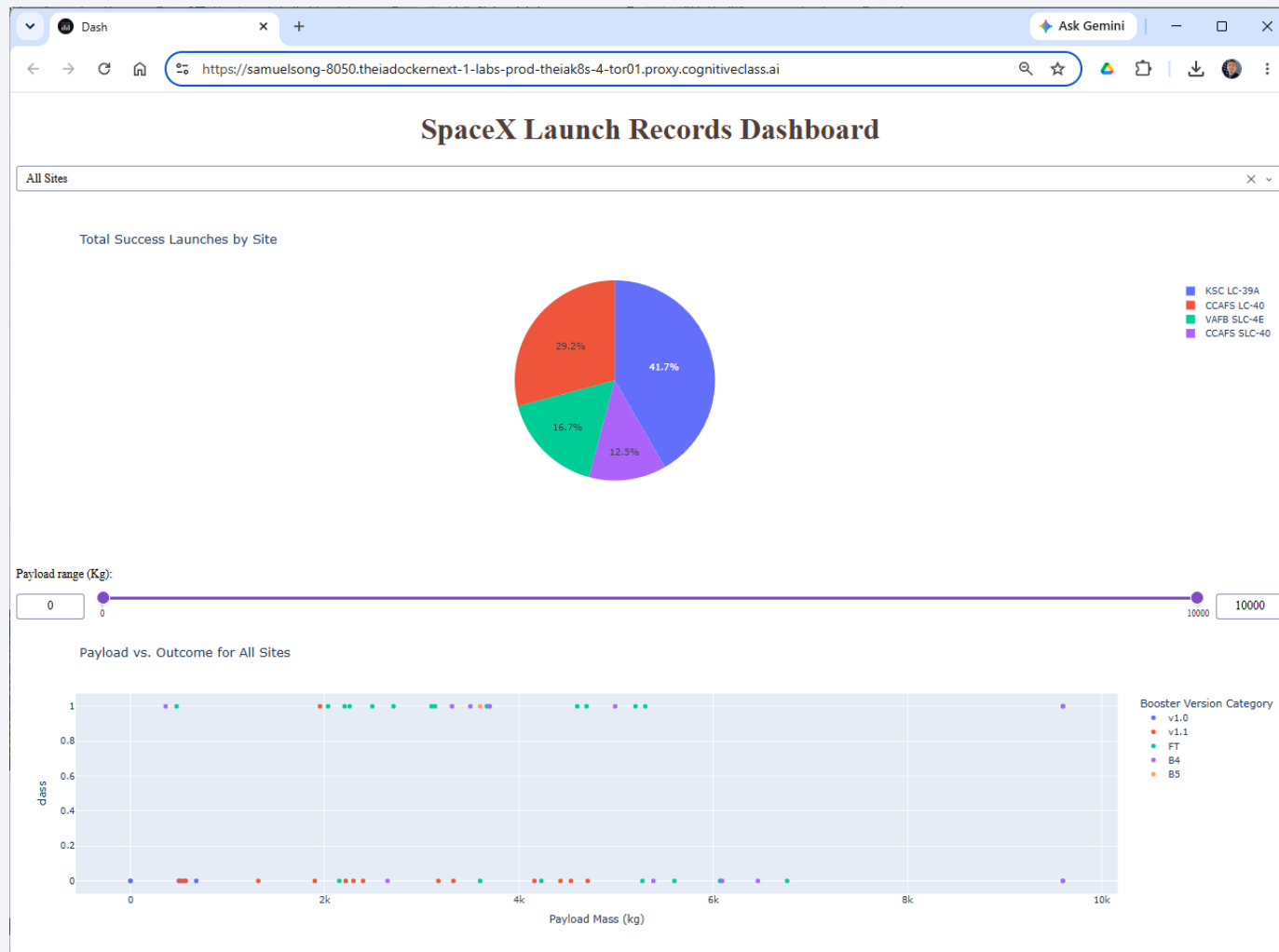
Launch Site with the Highest Success Rate



KSC LC-39A

- Class 1 Successful Launches: 76.9%
- Class 2 Unsuccessful Launches: 23.1%
- KSC LC-39A launch site is reliable

Payload vs. Launch Outcome Scatter Plot



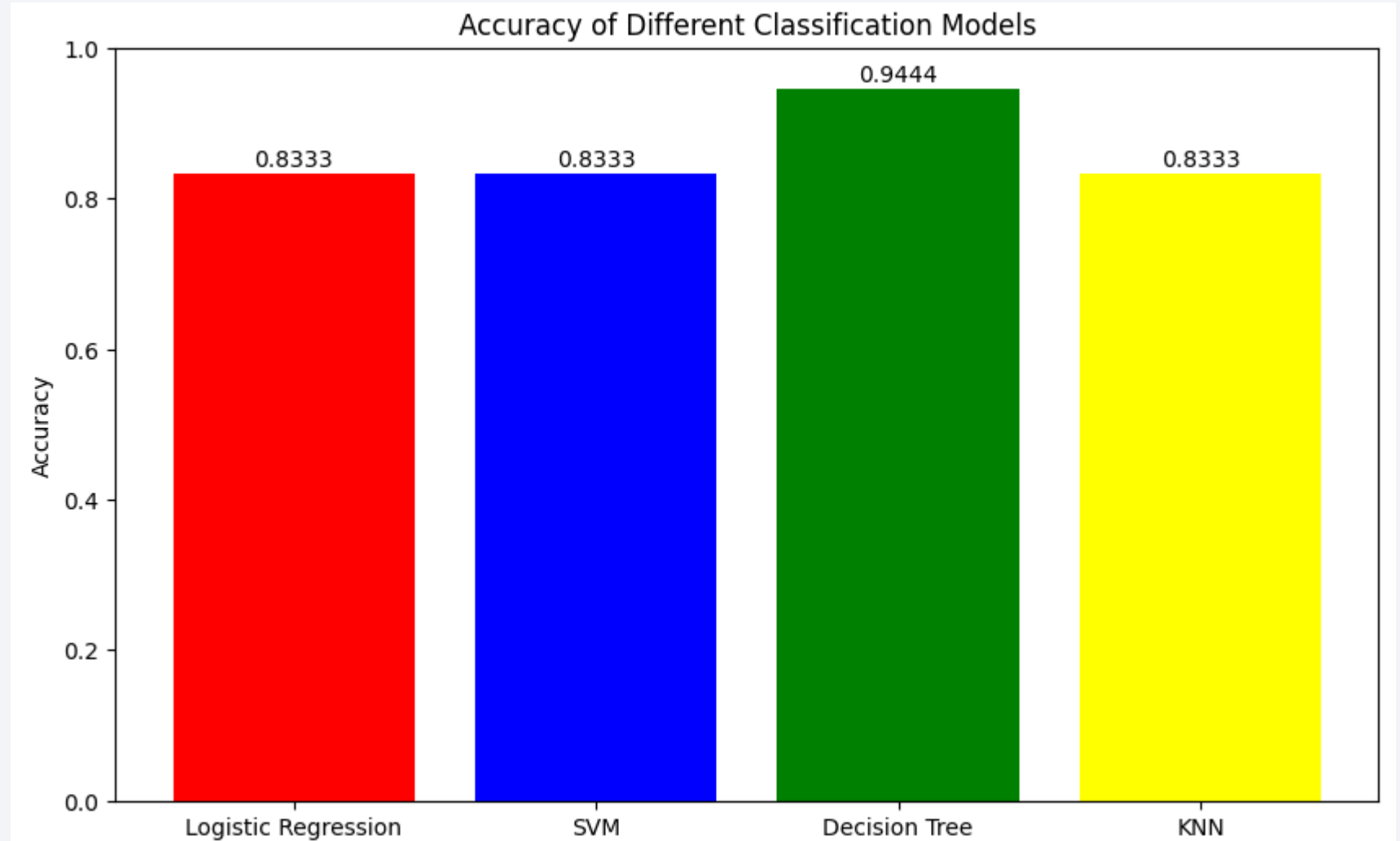
- CCAFS LC-40 has the highest success rate of successful launches proving it the most reliable.
- KSC LC-39A, VAFB SLC-4E, and CCAFS SLC-40 have lower success rates.
- Booster version “FT” was the most used with a high success rate.
- Booster version “v1.0” was used the least.

Section 5

Predictive Analysis (Classification)

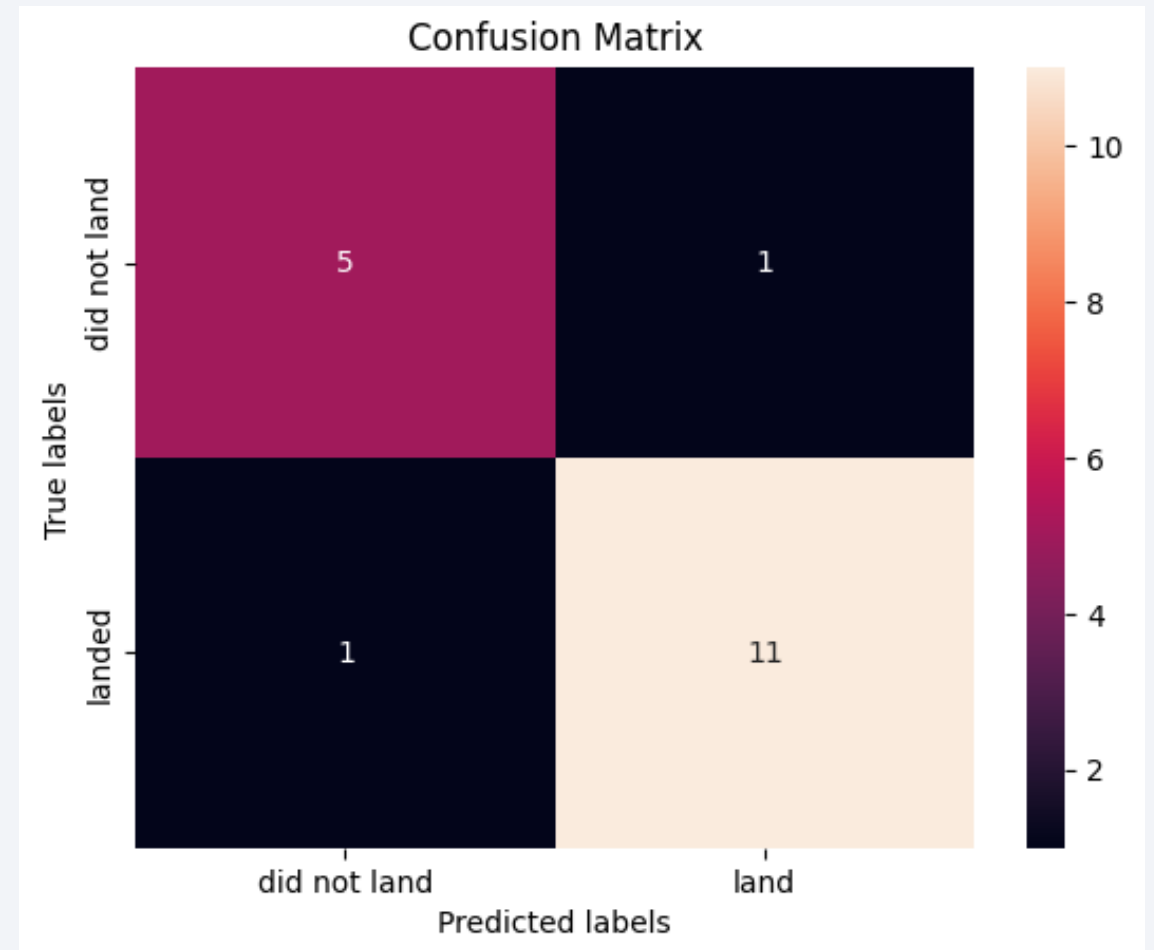
Classification Accuracy

- The Decision Tree model has the highest classification accuracy on the test data, achieving an accuracy of 0.9444. This suggests that the Decision Tree model is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors, all of which achieved an accuracy of 0.83



Confusion Matrix

- Decision Tree model had the highest accuracy score of 94.4%.
- True Positive of 5 and True Negative of 11 which is a balanced performance.
- No False Negatives
- 1 False Positive which is manageable.



Conclusions

- Decision Tree is the best model for predicting landing success
- There were more launches with a low payload than larger payload.
- Success rates of the launches increase year by year.
- KSC LC-39A has the highest success rate of the launches.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

